



New Stars Of Data
2026-05-08

Mastering Power BI Model Types

Import-, DirectQuery- or
Composite models?



Who am I

Or who makes that noise...

- Gregor Prohaska
- Operations Manager @ COSMO CONSULT
- „Some“ years of experience

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- Father of 2
- Curious by nature
- Don't take myself too seriously ;-)



01 Terms and Concept

02 Base Models

03 Composite Models

04 Which one to use...

05 Link to slide-deck 😊



Terms and Concepts

Source
Groups &
Relationships

Analysis
Services
engine

Storage
Modes

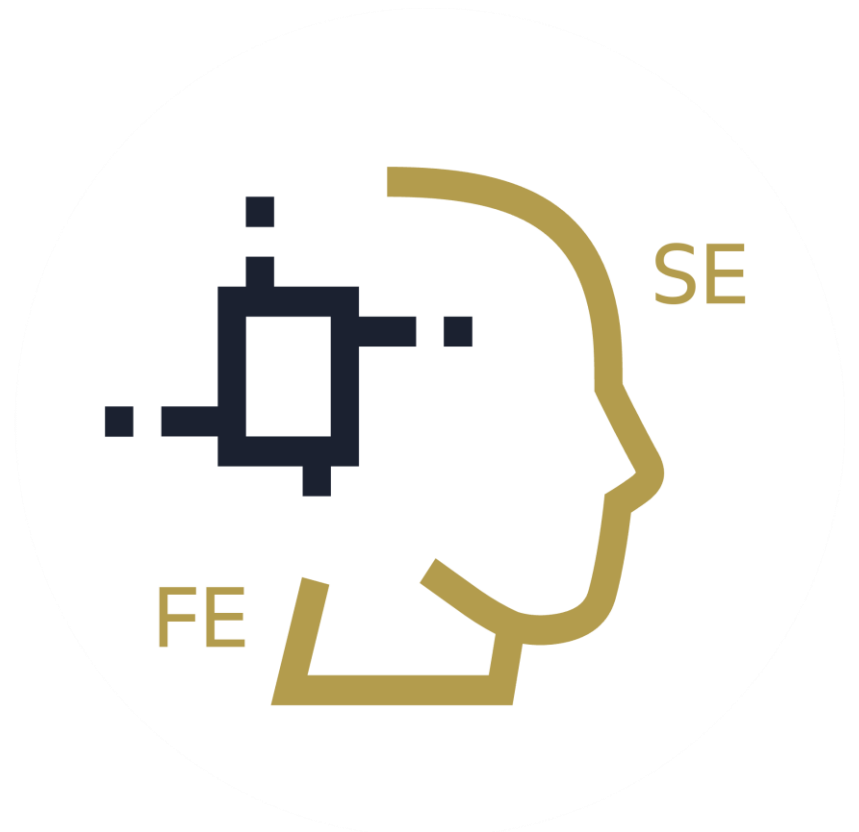
Semantic
Model

Performance



Analysis Service Engine

- Query-Execution Engine
- Formula Engine (FE) generates query plan and solves complex tasks
- Storage Engine (SE) retrieves data from data store



Further reading: <https://www.sqlbi.com/articles/formula-engine-and-storage-engine-in-dax/>



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Semantic
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Performance

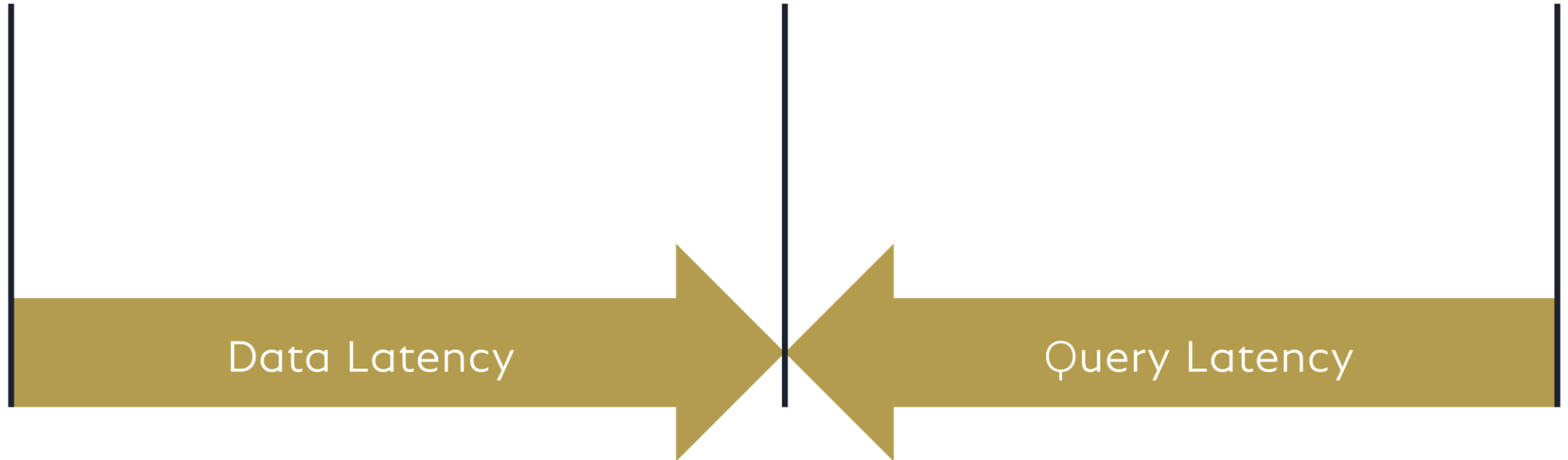


Performance

Data creation

Data available

Answer



Terms and Concepts

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Storage
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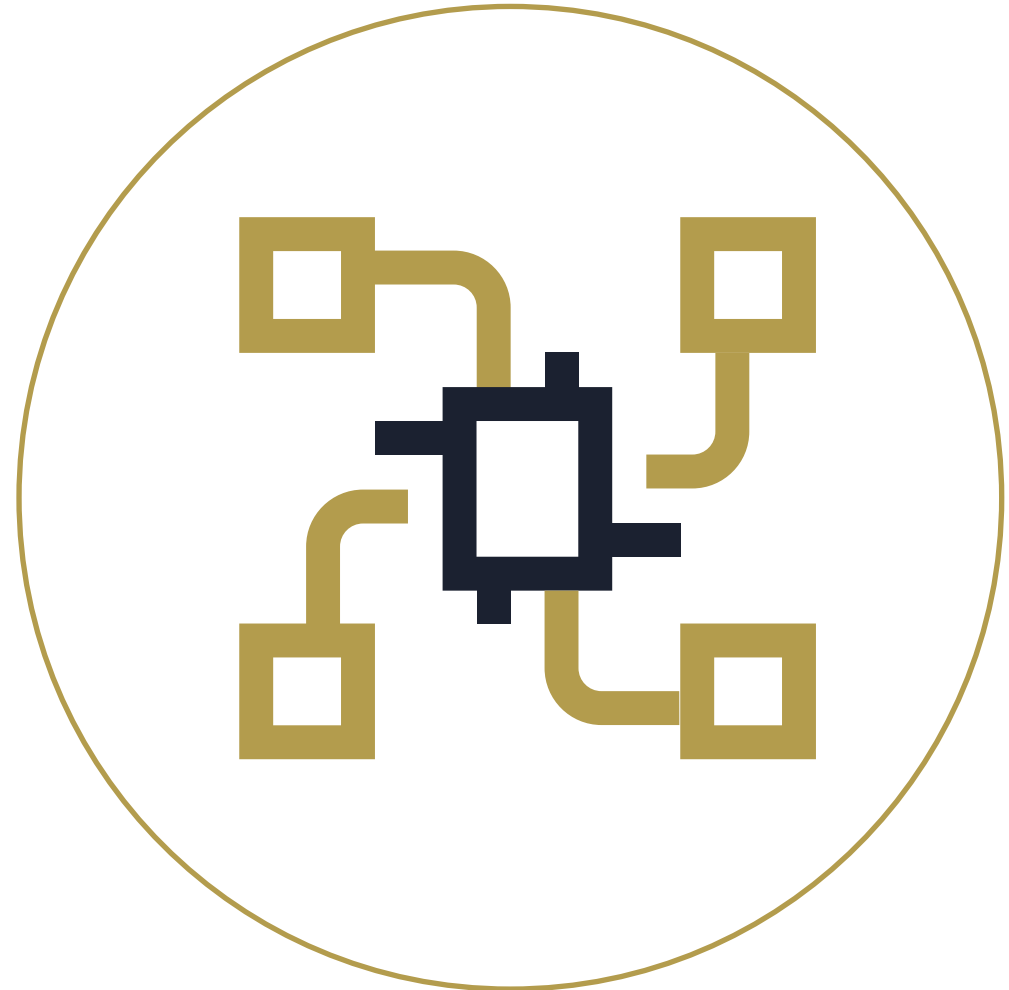
Semantic
Model



Semantic Model

(formerly know as Datasets)

- Logical layer
- Defines tables & their relationships
- Provides metadata such as formating & aggregations



Terms and Concepts

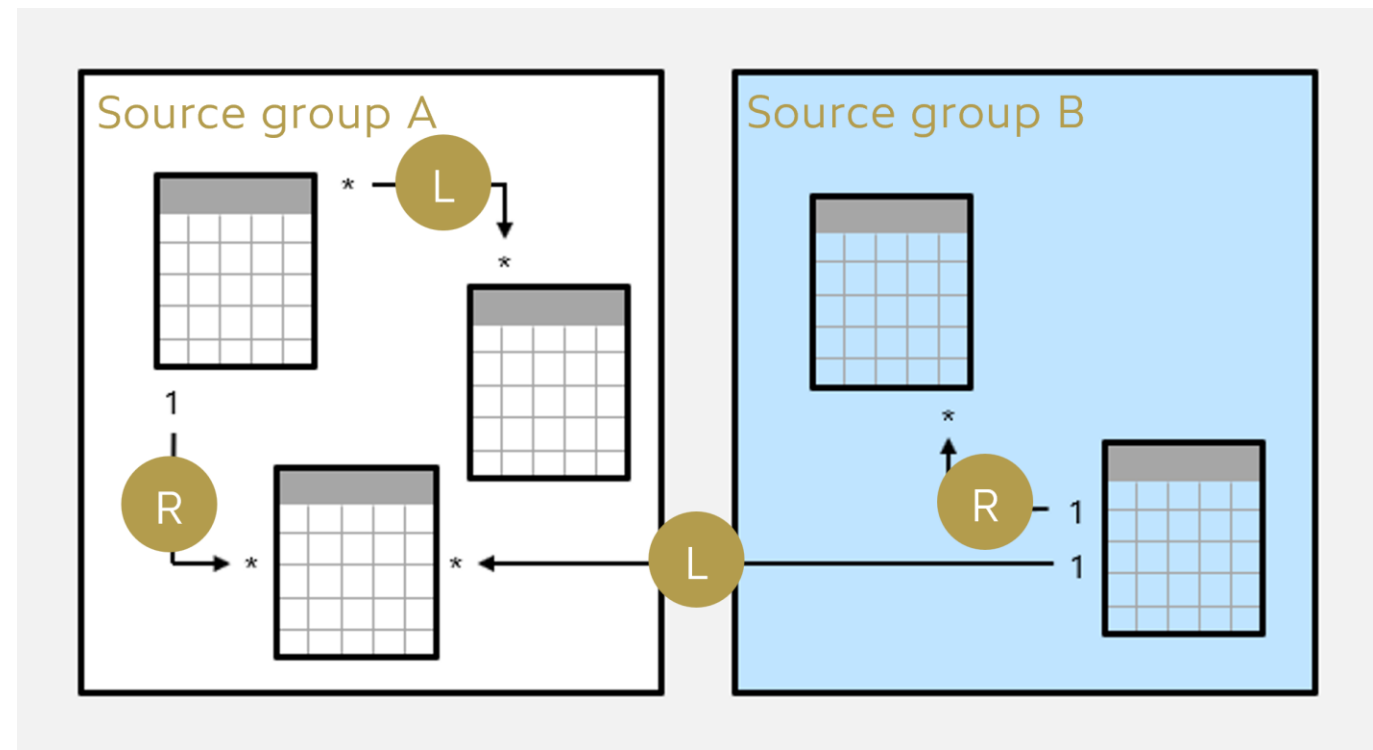
Source
Groups &
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Storage
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Source Groups & Relationships

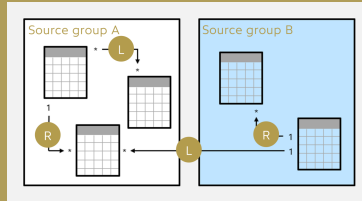
- Each Data Source is a source group
- Regular relationships are created for 1-N relations within a source group
- Limited relationships for M-N or cross source group relations



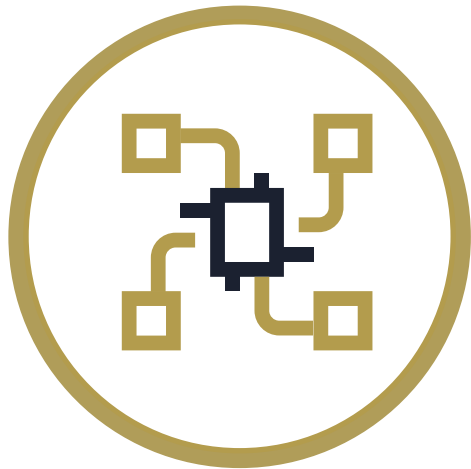
adapted from <https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-relationships-understand#relationship-evaluation>



Terms and Concepts

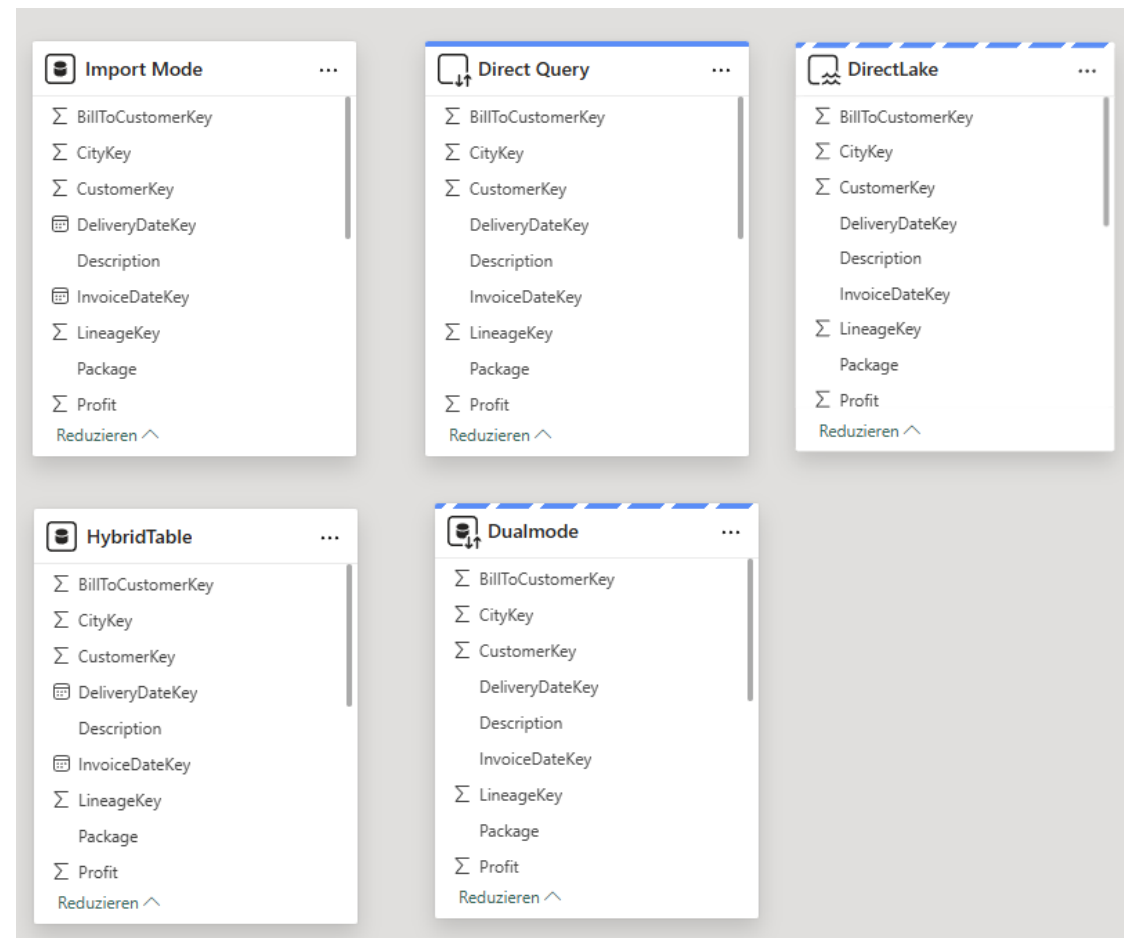


Storage
Modes

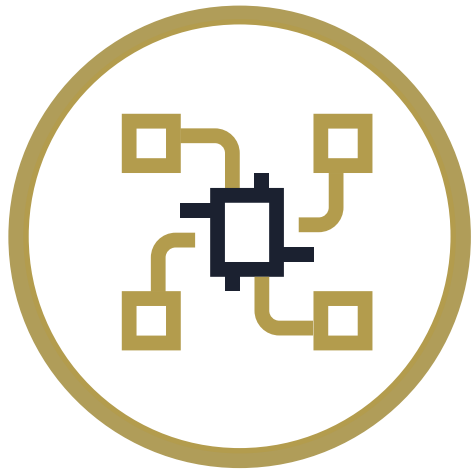
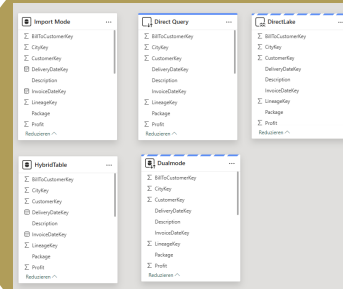
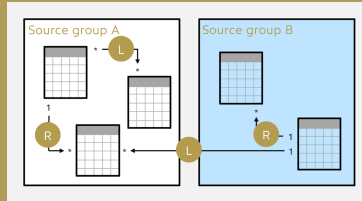


Storage Modes

- Defined per table
- 2 base types, 2 variations & Direct Lake



Terms and Concepts

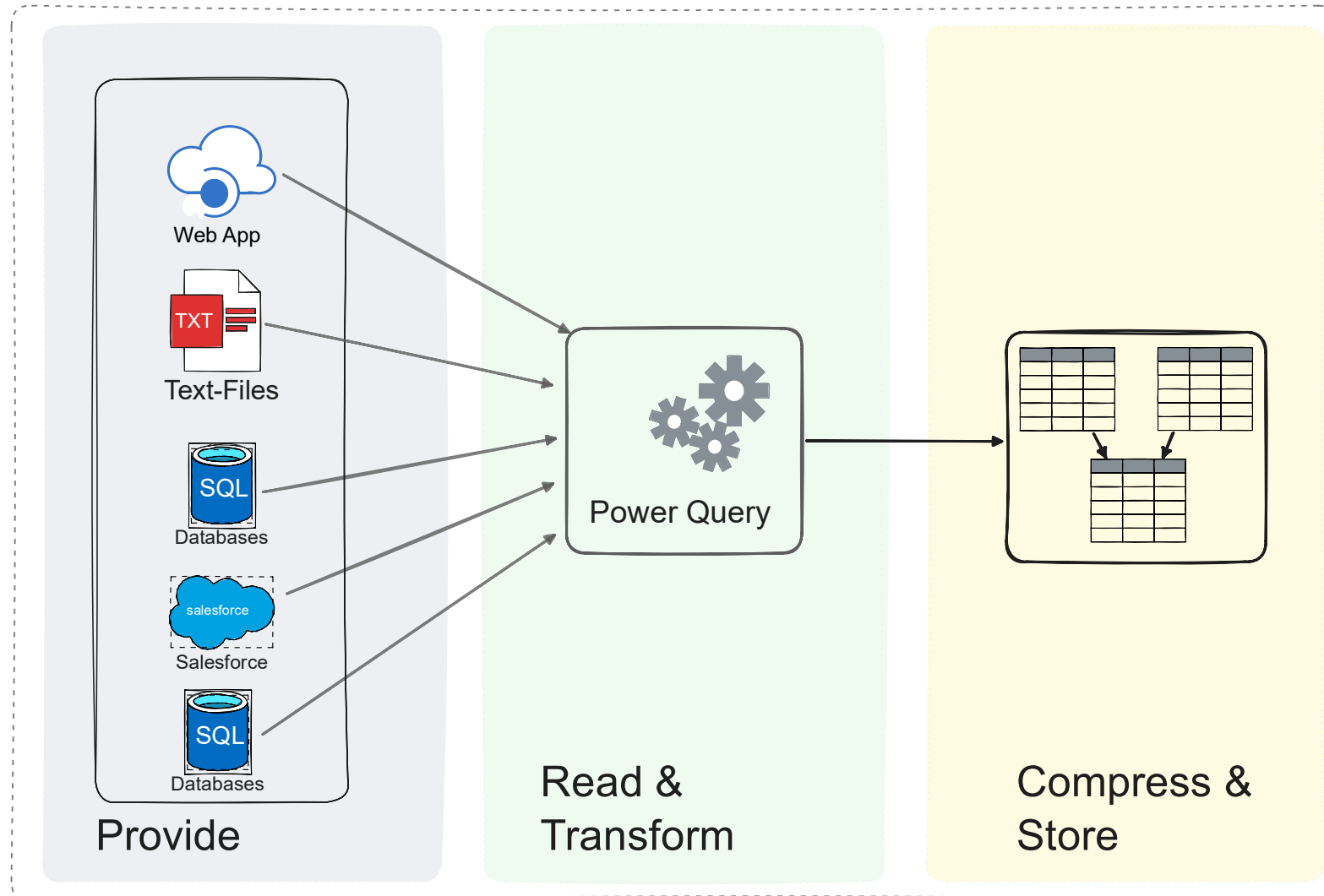


Base Models

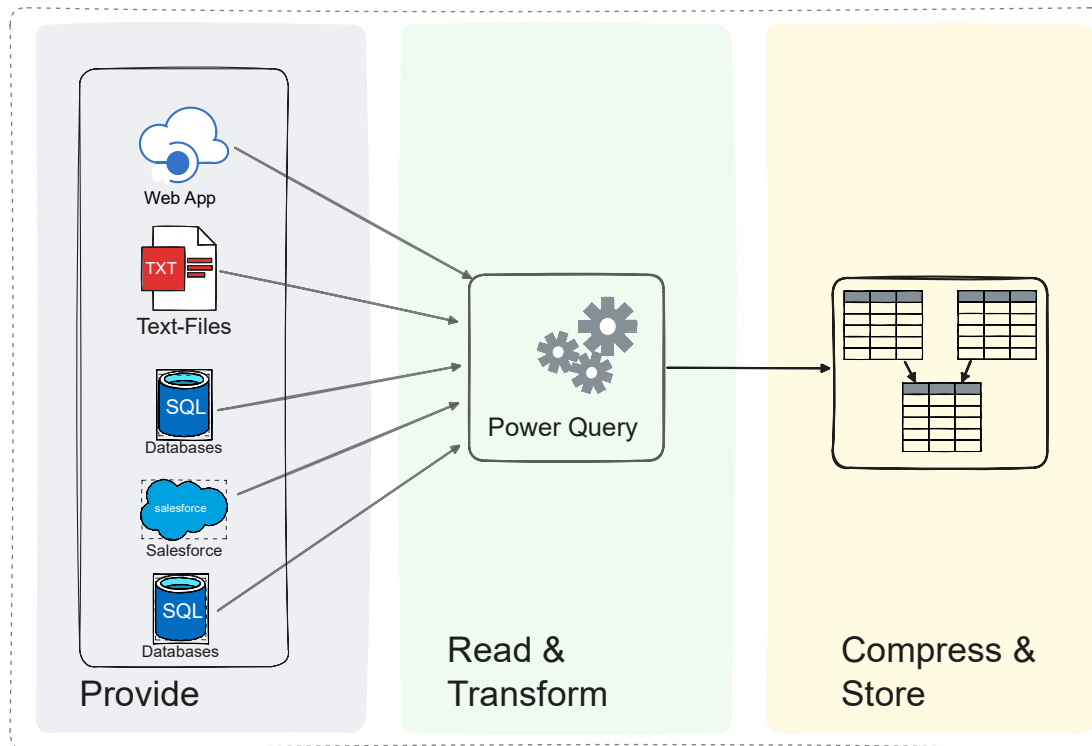
1. Import Mode
2. DirectQuery Mode
3. Direct Lake Mode
on OneLake
on SQL Analytics Endpoint



Import Model



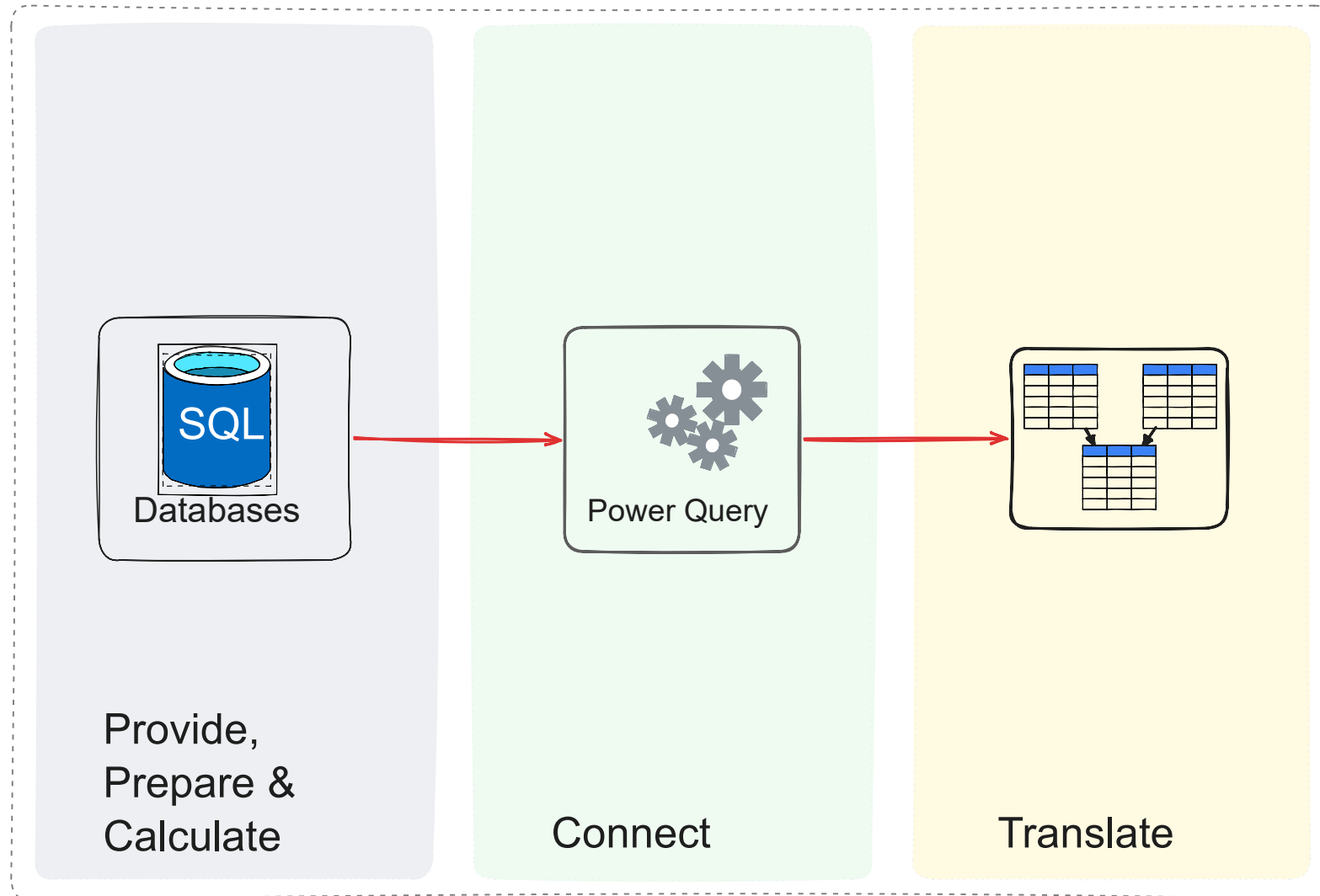
Import Model



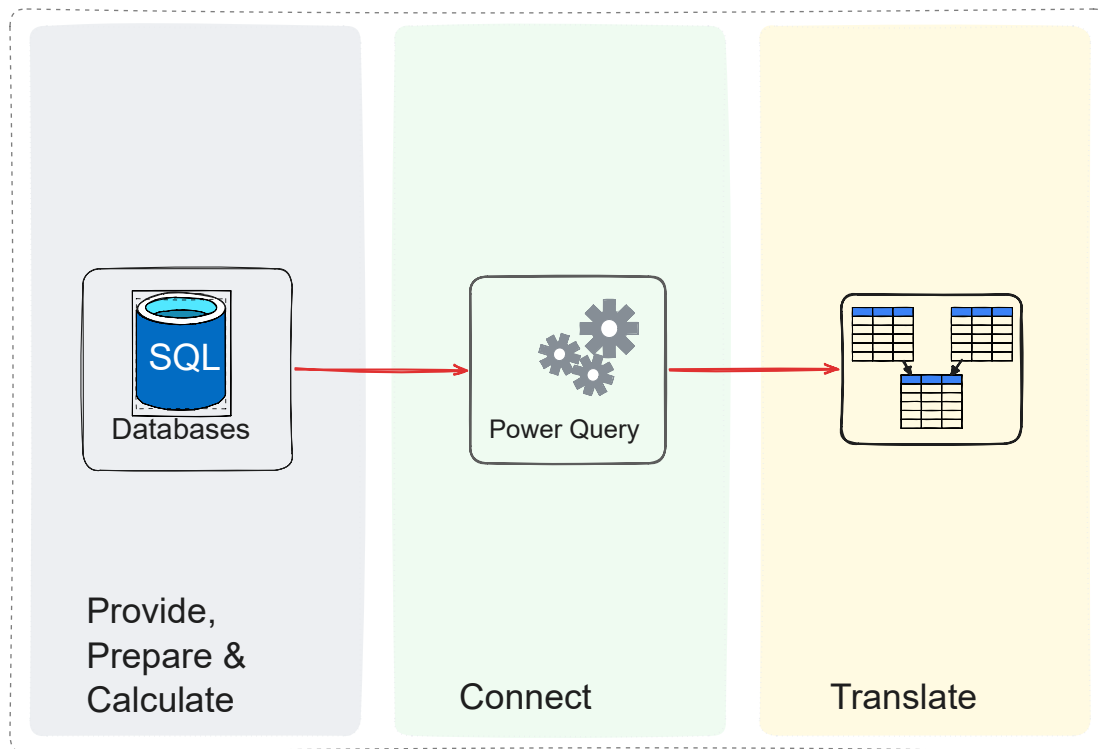
- Supported by all connectors/data sources
- Vertipaq engine
- Decouples import and query workload
- Semantic model security only
- Limited storage & (scheduled) refreshes



DirectQuery Model



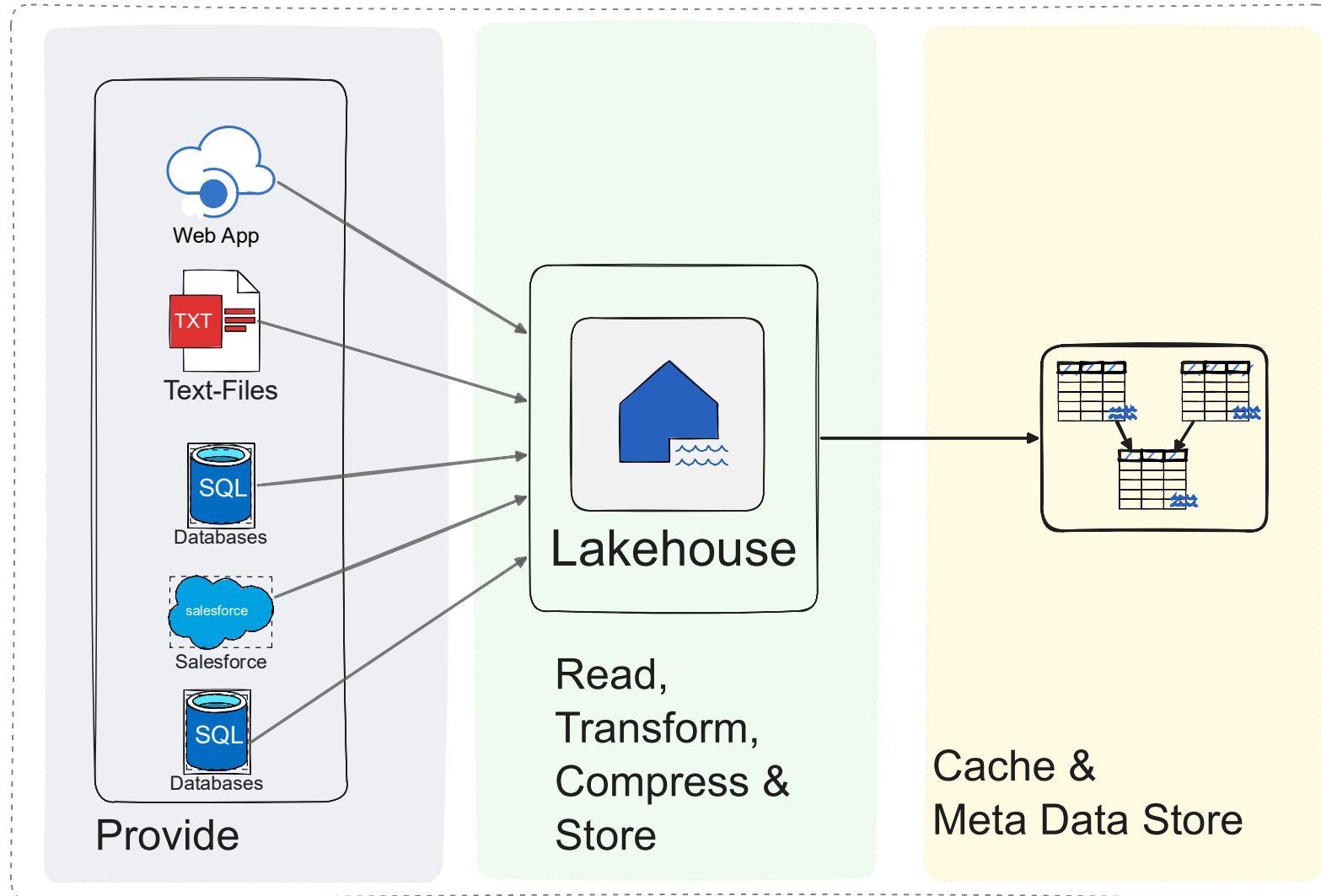
DirectQuery Model



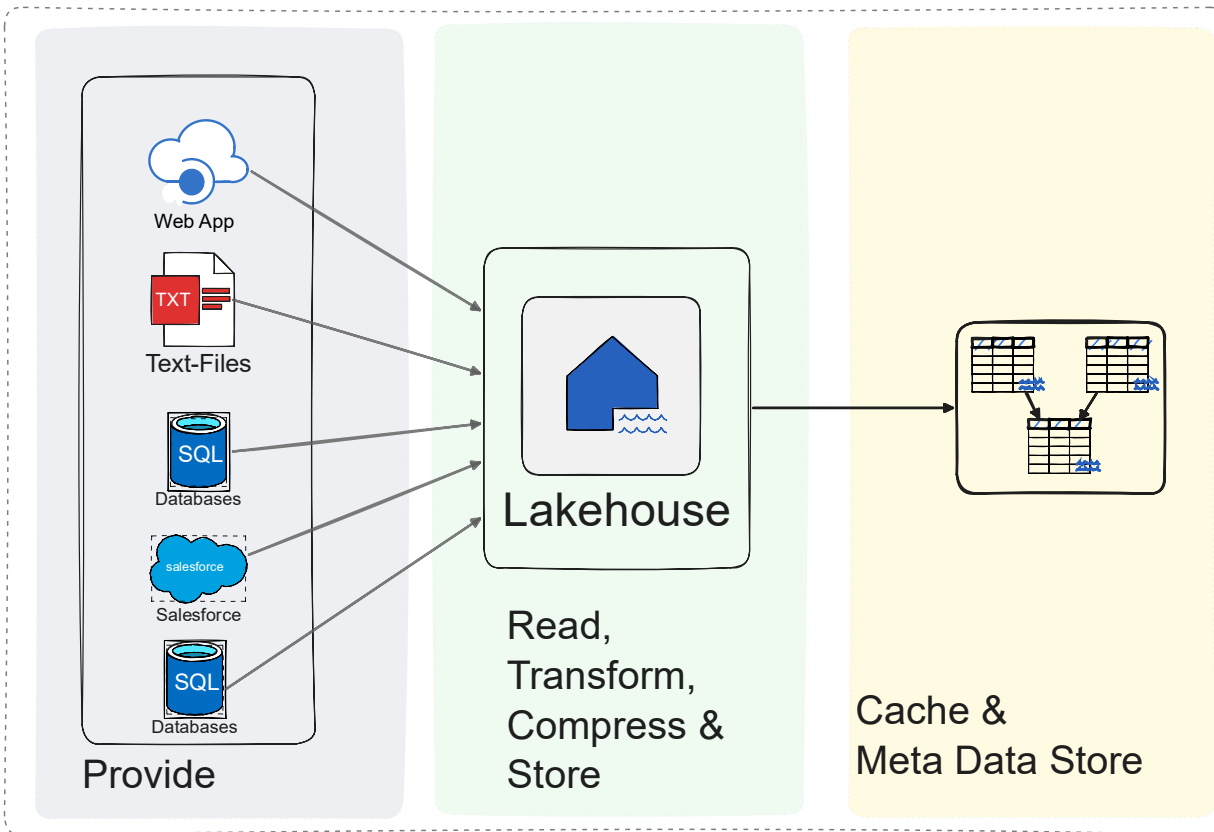
- Data stays at source & always up-to-date
- „no“ model size limits
- Data source security can be leveraged
- Limited DAX & data source support
- Calculated tables not supported



Direct Lake Model



Direct Lake Model



- Vertipaq engine
- Decouples import and query workload
- Model size limited by fabric capacity
- Data source security can be leveraged
- Limited support on calculated table

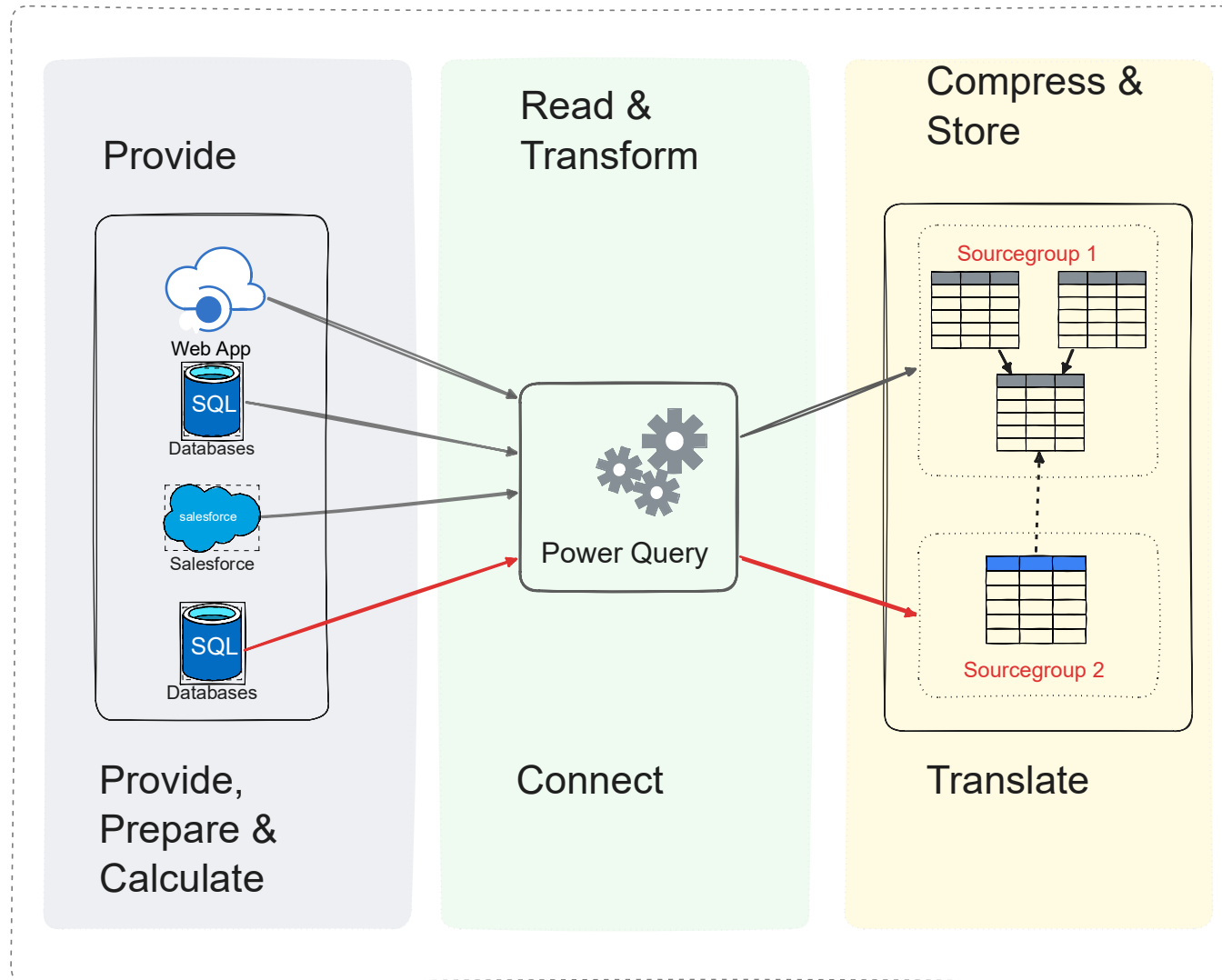


Composite Models

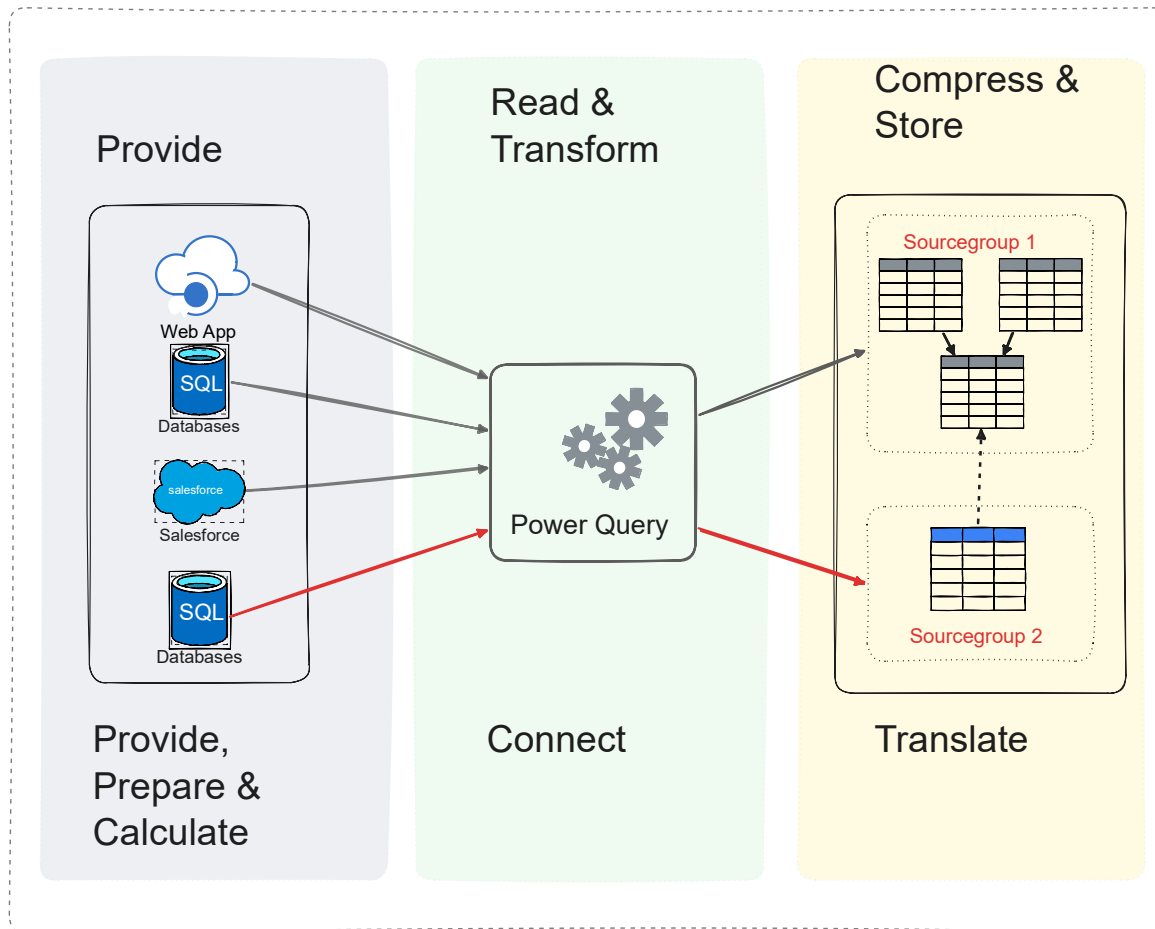
1. Import & DirectQuery
2. Multi-DirectQuery Mode
3. Import & Direct Lake
4. DirectQuery for Power BI



Composite Model – Import + DirectQuery



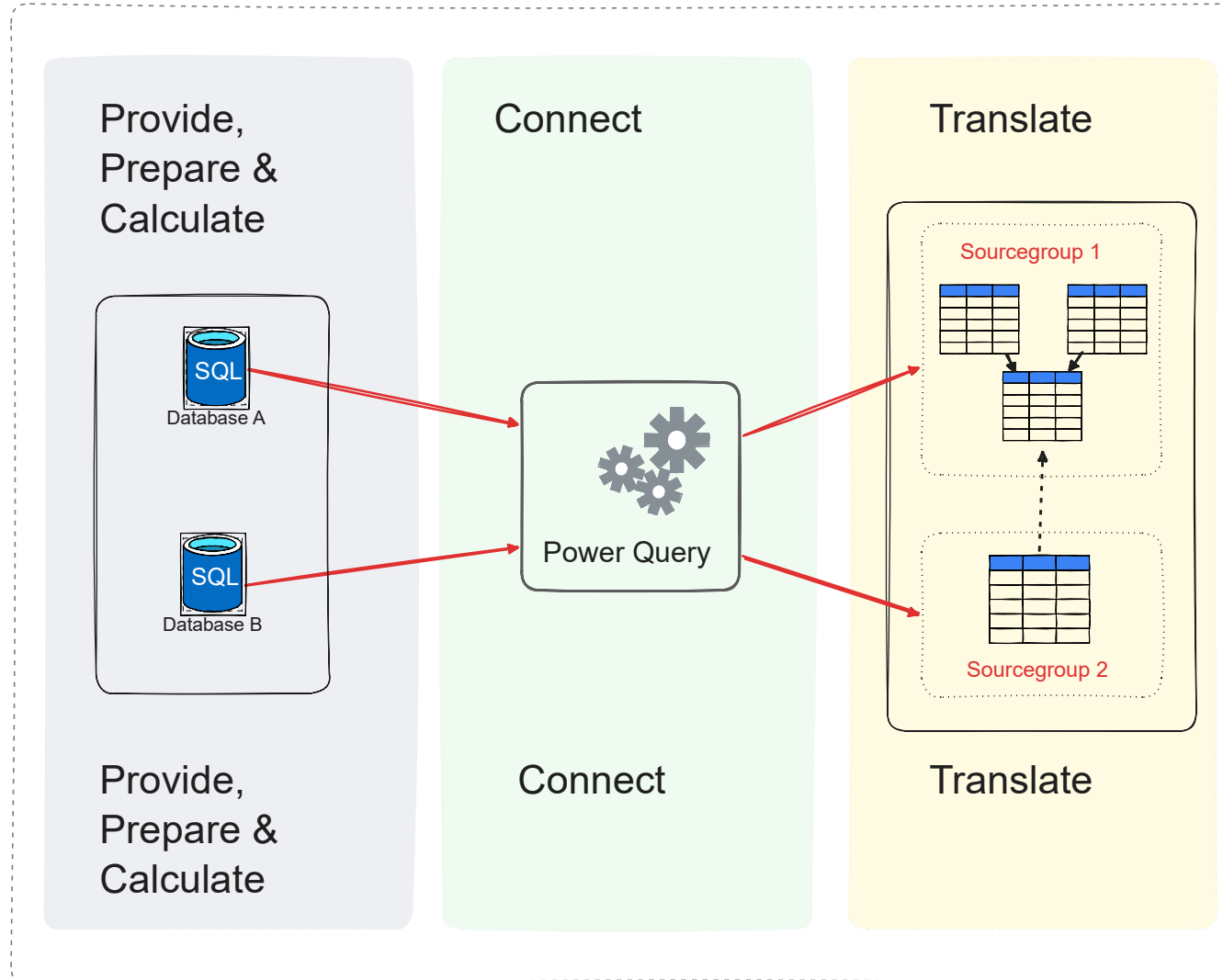
Composite Model – Import + DirectQuery



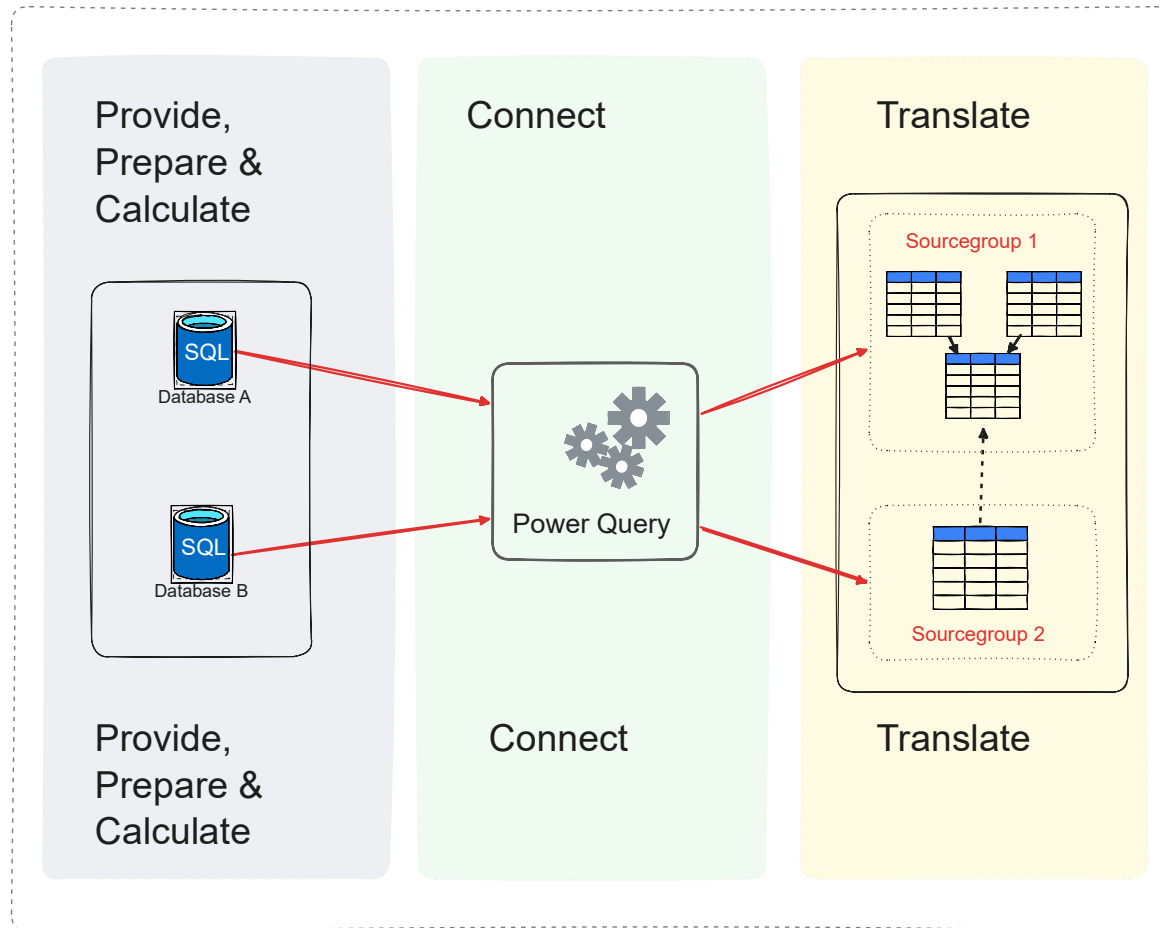
- Import and DirectQuery-Tables
- Dual-Mode possible (regular relationships)
- Limitations of DirectQuery-models apply (DQ only)
- Data source security can be leveraged (DQ only)



Composite Model – Multi-DirectQuery



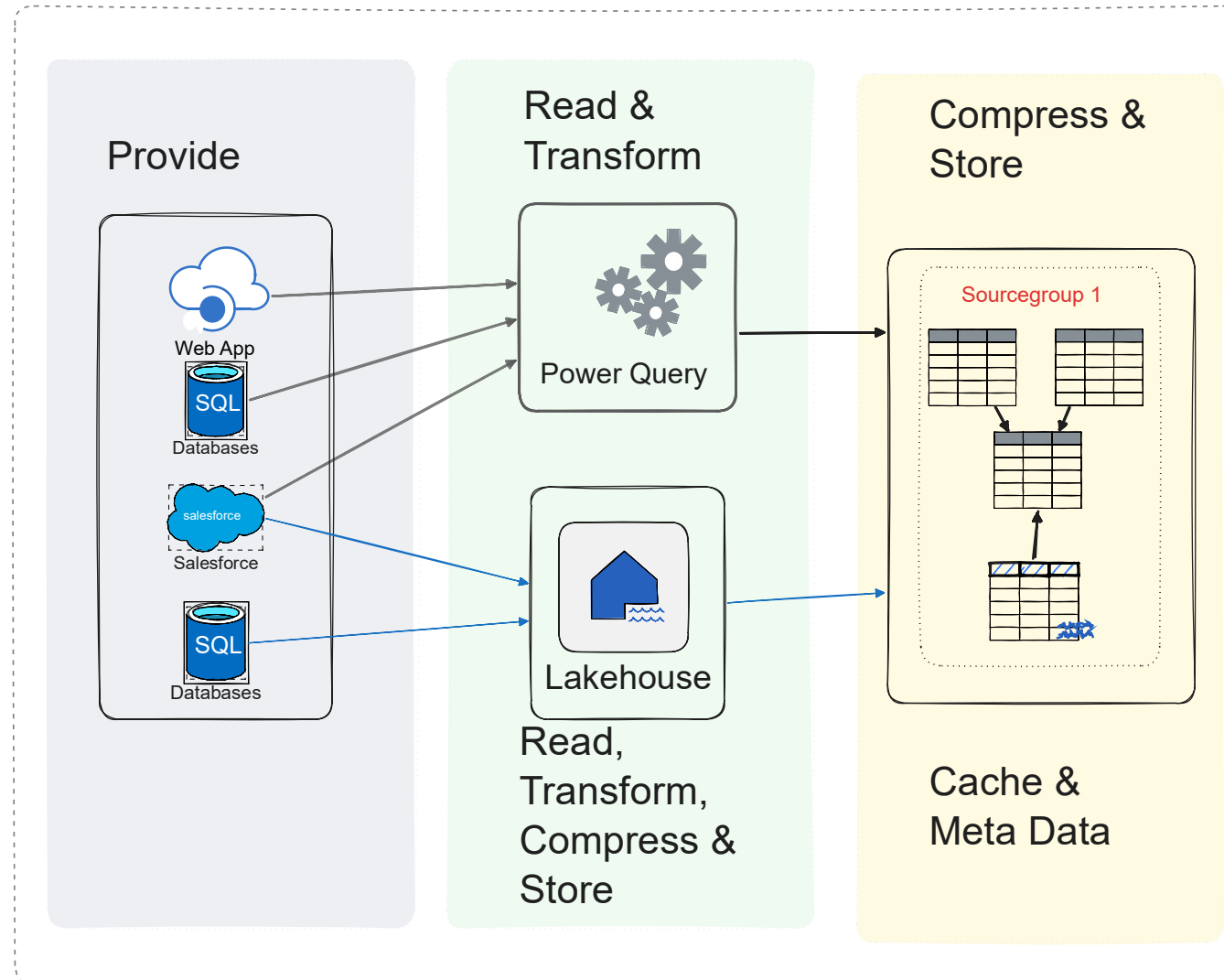
Composite Model – Multi-DirectQuery



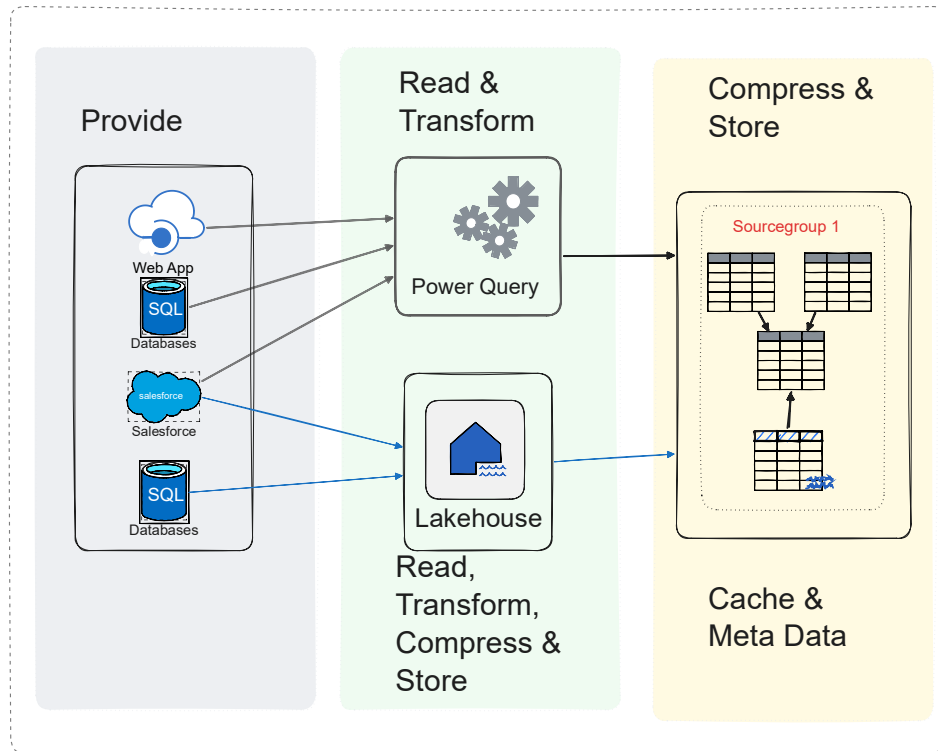
- Tables from different DirectQuery sources
- Limitations of DirectQuery-models apply
- Data source security can be leveraged
- Consider linking data sources upstream



Composite Model – Import + Direct Lake on OneLake



Composite Model – Import + Direct Lake on OneLake

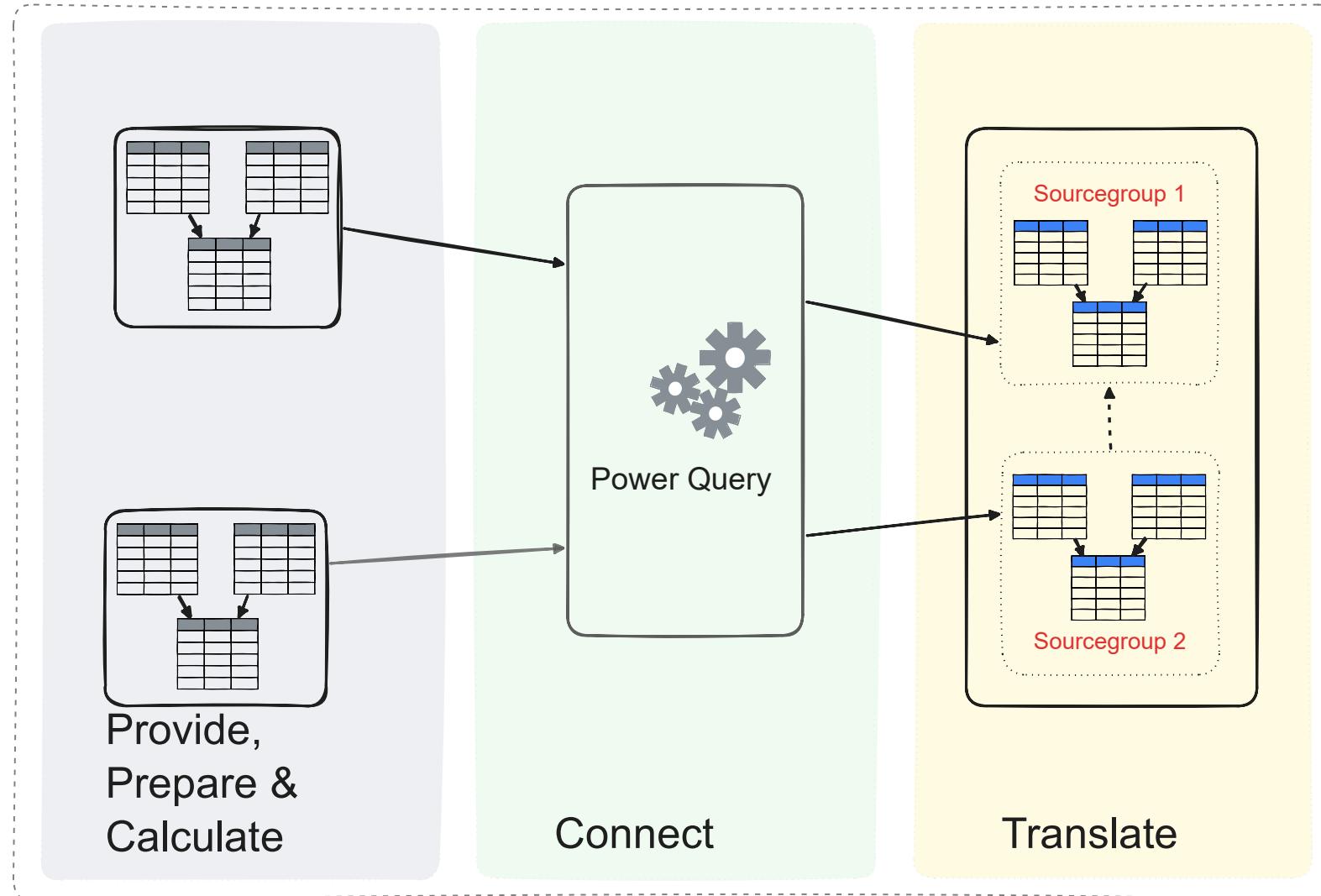


- Contains Import and Direct Lake Tables
- One Sourcegroup
- Improved performance
- All limitations of Direct Lake apply (DL only)



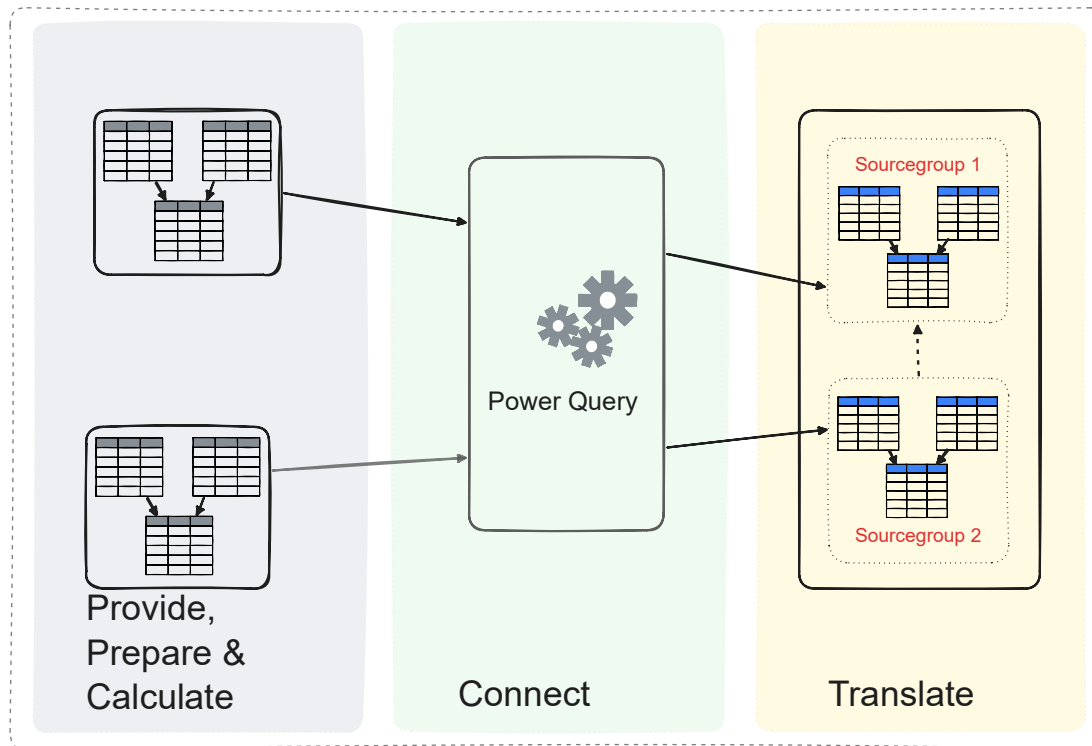
Composite Model – Direct Query for Power BI

... and Analysis Services 2022+



Composite Model – Direct Query for Power BI

... and Analysis Services 2022+



- Multiple semantic models can be combined in one
- Further Data can be added (Import and DQ)
- Source semantic model security is applied
- Consumers need READ-Permisson on all models



Storage Mode Cheat-Sheet

	Import	DirectQuery	Direct Lake
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Storage Mode Cheat-Sheet

	Import	DirectQuery	Direct Lake
Security	semantic model	data source or semantic model	Data source (Fabric) or semantic model



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Workspace types	Pro, PPU, Fabric	Pro, PPU, Fabric	Fabric



Which one to use...

- DirectQuery
 - Best for data freshness on relational or multi-dimensional data sources at the costs of query performance
- Direct Lake
 - Best for data freshness if data resides in OneLake already. Use it for irregular & rarely queried models in Fabric
- Composite Model – Import & DirectQuery
 - Use it for planning/write-back scenarios & if you need aggregations to overcome size limitations in non-fabric environments
- Composite Model – Import & Direct Lake
 - Use it if you need flexibility on dimensions (e.g. calculated columns) combined with huge fact tables
- Composite Model – DirectQuery on PBI ★
 - Use it for cross model reporting to enhance maintainability or to enable self-service enrichment
- Import Mode
 - Best in query performance, most flexible solution, use it as long as you don't have strong arguments against it



Thank you,

stay curious &
please give feedback



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Slides: <https://bit.ly/PowerBIModeltypes202605108>